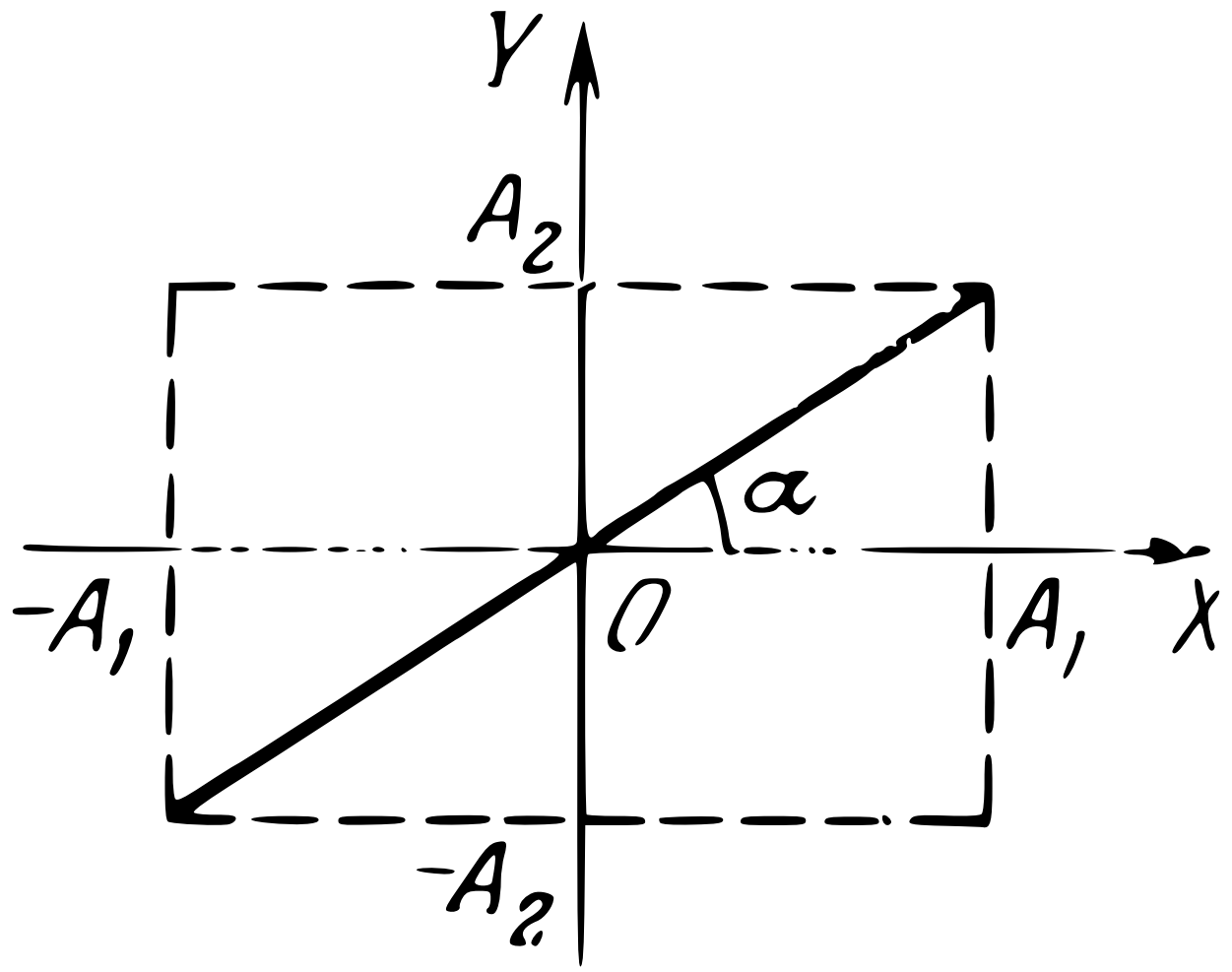
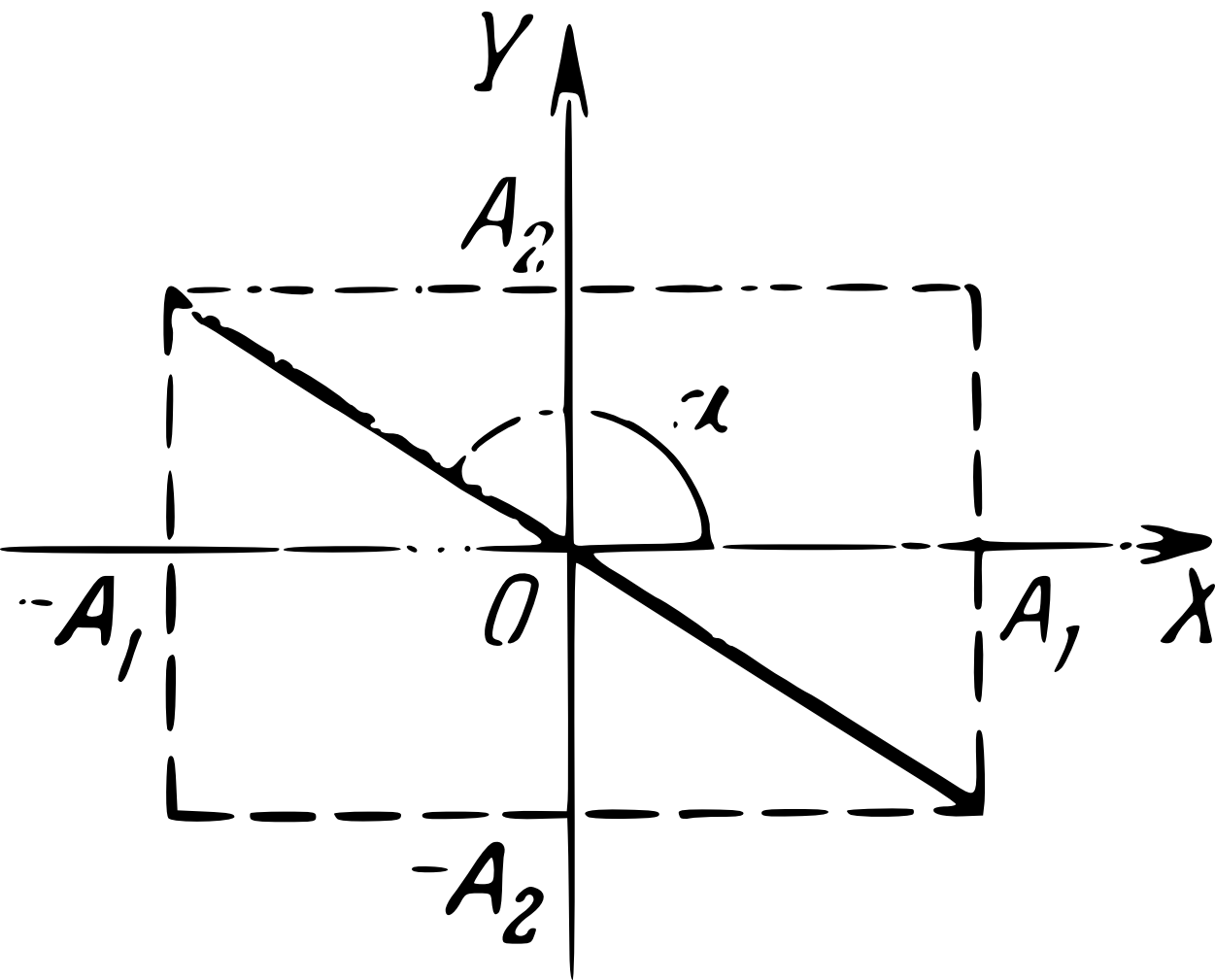
(a) $m = 0, \pm 2, \pm 4, \ldots$



(b) $m = \pm 1, \pm 3, \ldots$